Cambridge O Level Chemistry 5070

Ion tests and gas tests
Question paper extracts + mark schemes

2015 – 2025 · one question per test
Paper 2, plus Paper 3 only if Paper 2 has no write-in
Colours inverted.

Cl^- Br^- I^- SO_4^{2-} NO_3^- NH_4^+/NH_3 CO_2 SO_2 Cl_2 H_2 O_2
Fe^{2+}/Fe^{3+} Cu^{2+} Zn^{2+} Al^{3+} Cr^{3+}
Flame Na yellow K lilac Cu blue-green

Contents

2015–2025 · inverted extracts

(iii) Describe a test for chloride ions.

test ....

observation ....

[2]

[Total: 13]

Question	Answer	Marks
2(e)(ii)	*anode:* chlorine **AND** *cathode:* hydrogen	**1**
2(e)(iii)	(add nitric acid) then (aqueous) silver nitrate (1) white precipitate (1)	**2**

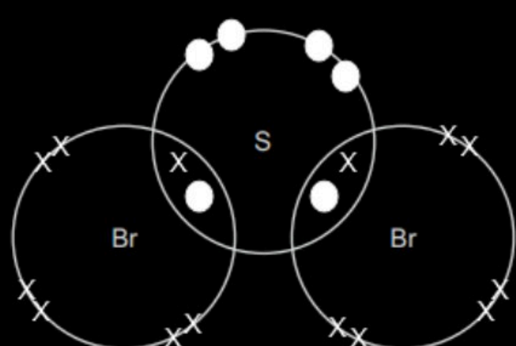
Question	Answer	Marks
3(a)	**Any three from:** • (property on which distillation) depends is the boiling point / copper(II) sulfate has higher boiling point than water / ORA (1) • idea of distillation apparatus, e.g. flask connected to condenser (1) • flask or solution heated (1) • idea that only water vaporised (when flask heated) (1) • water vapour converted to (liquid) water (in condenser) (1)	**3**
3(b)	filtration	**1**
3(c)	Cu $\frac{21.09}{64}$ Cs $\frac{43.82}{133}$ C*l* $\frac{35.09}{35.5}$ **OR** Cu = 0.33 Cs = 0.33 C*l* = 0.99 (1) $CuCsCl_3$ (1)	**2**

(c) Describe a test for aqueous bromide ions. Include the observations for a positive result.

test ....

observations ....

Question	Answer	Marks
3(d)	the particles move further apart	1

Question	Answer	Marks
4(a)	anode: bromine (1) cathode: hydrogen (1)	2
4(b)	anode: $2Br^- \rightarrow Br_2 + 2e^-$ (1) cathode: $Ca^{2+} + 2e^- \rightarrow Ca$ (1)	2
4(c)	add (acidified) silver nitrate (1) cream precipitate (1)	2
4(d)(i)	calcium gives electrons to zinc <u>ions</u> / electrons transferred from calcium to zinc <u>ions</u>	1
4(d)(ii)	+2	1
4(e)	 pair of electrons between each S and Br (1) 6 non-bonding e^- on each Br and 4 non-bonding e^- on S (1)	2

(c) Describe a chemical test for the iodide ion.

test ....

observation ....

.... [2]

Question	Answer	Marks
7(a)	ammonium iodide disappears / solid disappears / sample disappears / nothing left in tube (1)	**1**
7(b)	moles of ammonium iodide $\frac{2.9}{145}$ **OR** 0.02(00) (1) moles of gas = 2 × 0.02 **OR** 0.04(0) (1) volume of gas = (0.04 × 24) = 0.96 dm^3 **OR** 960 cm^3 (1)	**3**
7(c)	(acidified aqueous) silver nitrate (1) (pale) yellow ppt (1)	**2**
7(d)	$2I^-(aq) + Br_2(aq) \rightarrow I_2(aq) + 2Br^-(aq)$ correct formulae and balanced (1) correct state symbols – dependent on correct formulae (1)	**2**
7(e)	in solid ions cannot move (1) in aqueous / solution ions can move (1)	**2**

(b) Describe a test for sulfate ions.

test ....

result ....
[2]

Question	Answer	Marks
B7(a)(i)	Magnesium has strong bonding between positive ions / cations and electrons / magnesium is a giant structure (1) Sulfur is a simple molecule / weak forces between sulfur molecules (1)	**2**
B7(a)(ii)	Magnesium has electrons which move (from place to place) (1) Sulfur does not have delocalised electrons / no mobile electrons / electrons don't move (1)	**2**
B7(b)	Giant structure / many covalent <u>bonds</u> (1) Need high temperature / lot of energy to break the <u>bonds</u> (1)	**2**
B7(c)(i)	Mass of sulfur = 19.2 g (1) mol S = $\frac{19.2}{32}$ mol C*l* = $\frac{21.3}{35.5}$ **OR** ratio = 0.6 to 0.6 (1) SC*l* (1)	**3**
B7(c)(ii)	S_2Cl_2 (1)	**1**

Question	Answer	Marks
B8(a)	$\frac{2 \times 39}{174} \times 100$ = 44.8% / 45% (2 marks) If 2 marks not scored correct M_r = 174 (1)	**2**
B8(b)	(Acidified) barium chloride / barium nitrate (1) White precipitate (1)	**2**
B8(c)	Nitrates soluble (in water) / nitrates dissolve (easily) (1)	**1**

(c) reacts with warm aqueous sodium hydroxide and aluminium foil to give a gas that turns damp red litmus paper blue

.... [1]

(d) has an aqueous solution that is used to test for an oxidising agent

.... [1]

(e) has an aqueous solution that reacts with copper metal.

.... [1]

[Total: 5]

Question	Answer	Marks
1(a)	barium sulfate	**1**
1(b)	copper(II) sulfate	**1**
1(c)	silver nitrate	**1**
1(d)	potassium iodide	**1**
1(e)	silver nitrate	**1**

Question	Answer	Marks
2(a)(i)	each carbon atom in the structure uses three outer shell electrons in forming covalent bonds (1) presence of mobile electrons (between layers) (1)	**2**
2(a)(ii)	inert	**1**
2(b)	(anode) bromine / Br_2 (1) (cathode) copper / Cu (1)	**2**
2(c)	$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$ / $4OH^- - 4e^- \rightarrow O_2 + 2H_2O$ (1) $2H^+ + 2e^- \rightarrow H_2$ (1)	**2**
2(d)(i)	water	**1**
2(d)(ii)	idea of difficult to store the hydrogen under pressure / availability of hydrogen sources	**1**

(c) Describe a chemical test for the ammonium ion.

test ....

observation ....

....
[2]

(d) Aqueous ammonium carbonate reacts with dilute hydrochloric acid.

Question	Answer	Marks
7(a)	solid disappears / ammonium carbonate disappears / nothing left in tube (1)	**1**
7(b)	moles of ammonium carbonate = 4.80 / 96 **OR** 0.05(00) (1) moles of gas = 3 × 0.05 **OR** 0.15 (1) volume of gas = (0.15 × 24) = 3.6 dm^3 **OR** 3600 cm^3 (1)	**3**
7(c)	heat or warm with (aqueous) sodium hydroxide (1) gas that turns (moist red) litmus blue (1)	**2**
7(d)	$CO_3^{2-}(aq) + 2H^+(aq) \rightarrow CO_2(g) + H_2O(l)$ correct formulae (1) correct state symbols – dependent on correct formulae (1)	**2**
7(e)	in solid ions cannot move (1) in aqueous solution ions can move (1)	**2**

Question	Answer	Marks
8(a)	when a reversible reaction (in a closed system) (1) (reaches a point that the) rate of forward reaction equals the rate of the backward reaction (1)	**2**
8(b)	more PCl_5 / concentration of PCl_5 increases / less PCl_3 / less Cl_2 / concentration of Cl_2 decreases / concentration of PCl_3 decrease (1) there are fewer moles of gas on the left hand side of the reaction (1)	**2**
8(c)(i)	the reaction absorbs heat / the (forward) reaction is endothermic (1)	**1**

(b) Describe a chemical test for carbon dioxide.

test ....

observation if carbon dioxide present ....

Question	Answer	Marks
1(a)(i)	**A**	**1**
1(a)(ii)	**B**	**1**
1(a)(iii)	**F**	**1**
1(a)(iv)	**C**	**1**
1(a)(v)	**E**	**1**
1(b)	**C** and **D**	**1**

Question	Answer	Marks
2(a)	different number of neutrons / different mass number / different nucleon number	**1**
2(b)	same electronic configuration	**1**
2(c)	$\frac{(95.04 \times 32) + (0.75 \times 33) + (4.20 \times 34) + (0.01 \times 36)}{100}$ correct numerator (1) correct denominator (1)	**2**

Question	Answer	Marks
3(a)(i)	**A** activation energy / E_a (1) **B** enthalpy change / ΔH (1)	**2**
3(a)(ii)	product level above reactant level	**1**
3(b)	limewater goes milky	**1**

(ii) Describe the chemical test for sulfur dioxide.

test ....

....

Question	Answer	Marks
B10(a)(i)	Moles of Zn and H_2 = 0.01 (1) Volume of H_2 = 0.24 dm^3 (1)	2
B10(a)(ii)	The number of moles of iron is different (1)	1
B10(b)	**One mark each for any two observations from:** Solution gets warmer / heat given off (1) White precipitate formed (1) which redissolves in excess (1) Green precipitate formed (1) With excess the green precipitate remains (1) **One mark each for any two explanations from** Acid is neutralised (1) Green precipitate is iron(II) hydroxide (1) White precipitate of zinc hydroxide (1) which will redissolve in excess (1)	4
B10(c)(i)	Oxidation number of iron increases / oxidation number of iron becomes more positive / iron(II) ions lose electrons (1)	1
B10(c)(ii)	Use (filter paper dipped into acidified) potassium manganate(VII) (1) Purple colour changes to colourless (1)	2
	Total:	**10**

(c) Describe the chemical test for chlorine gas.

test ....

observation ....

Question	Answer	Marks
1(a)	hydrogen (1)	**1**
1(b)	sulfur dioxide (1)	**1**
1(c)	ethane (1)	**1**
1(d)	methane (1)	**1**
1(e)	ozone (1)	**1**
1(f)	carbon monoxide (1)	**1**

Question	Answer	Marks
2(a)	all have 7 electrons in their outer shell (1)	**1**
2(b)(i)	(colourless) to brown (solution) (1)	**1**
2(b)(ii)	iodide (ion) loses electrons (1)	**1**
2(b)(iii)	chlorine (molecule) gains electrons (1)	**1**
2(c)	test – (moist red or blue) litmus paper (1) observation – bleached / goes white (1)	**2**
2(d)(i)	rate increases particles are more crowded / more particles in the same volume / increased number of particles per unit volume (1) more collisions per second / increased collision frequency (1)	**2**
2(d)(ii)	alternative reaction pathway (1) has lower activation energy (1)	**2**

(b) Describe a chemical test for hydrogen.

test ....

observation if hydrogen present ....

Question	Answer	Marks
3(a)(i)	**A** activation (energy) / E_a (1) **B** enthalpy (change) / ΔH (1)	**2**
3(a)(ii)	product level below reactant level	**1**
3(b)	'pops' with a lighted splint	**1**
3(c)	(moles of hydrogen =) 366 / 24000 **OR** 0.01525 (1) (moles of Al =) 0.01525 × 0.67 **OR** 0.01017 (1) mass of Al = 0.27 (1)	**3**

Question	Answer	Marks
4(a)(i)	**Any two from**: **M1** random (arrangement) (1) **M2** (molecules are spread) far apart (1) **M3** (molecules are) moving (very) fast (1) **M4** random (motion)	**2**
4(a)(ii)	weak intermolecular forces	**1**
4(b)(i)	high melting point / high boiling point (1) conducts electricity in (aqueous) solution (1)	**2**
4(b)(ii)	$Li \rightarrow Li^+ + e^-$	**1**
4(b)(iii)	$Br_2 + 2e^- \rightarrow 2Br^-$	**1**

(d) Describe the test for oxygen.

test ....

observation ....

Question	Answer	Marks
1(a)	sulfur dioxide	**1**
1(b)	iron(II) oxide	**1**
1(c)	aluminium oxide	**1**
1(d)	calcium oxide	**1**
1(e)	sulfur dioxide	**1**

Question	Answer	Marks
2(a)	21(%)	**1**
2(b)	carbon dioxide is an acidic oxide (1) reacts with base / reacts with alkali / neutralised by sodium hydroxide (1)	**2**
2(c)	air liquefied (1) fractional distillation (1)	**2**
2(d)	glowing splint (1) relights (1)	**2**
2(e)(i)	photochemical / redox	**1**
2(e)(ii)	lightning	**1**
2(f)	(more) skin cancer / (more) sunburn / (more) harm to eyes	**1**

(d) Describe a chemical test to distinguish between aqueous iron(II) ions and aqueous iron(III) ions.

test ....

observations with aqueous iron(II) ions

Question	Answer	Marks
5(a)	magnesium loses electrons and is oxidation (1) iron(II) ions gain electrons and is reduction (1)	**2**
5(b)	(some of the) electrons move (throughout the structure)	**1**
5(c)	magnesium is too reactive / magnesium is very reactive / magnesium is high in the reactivity series	**1**
5(d)	add (aqueous) sodium hydroxide / (aqueous) ammonia (1) iron(II) ions give green precipitate (1) iron(III) ions give red-brown precipitate (1)	**3**
5(e)(i)	the ions can move	**1**
5(e)(ii)	*anode:* chlorine / Cl_2 (1) *cathode:* magnesium / Mg (1)	**2**
5(f)	does not corrode in water / does not corrode in air / unreactive	**1**

(b) Describe a test for copper(II) ions.

test ....

observations ....

Question	Answer	Marks
3(a)	(some of the) electrons can move / (some of the) electrons are mobile	**1**
3(b)	(aqueous) sodium hydroxide (1) (light) blue precipitate (insoluble in excess) (1) **OR** (aqueous) ammonia (1) (light) blue precipitate soluble in excess / dark blue solution formed in excess ammonia (1)	**2**
3(c)(i)	positive electrode: bubbles / effervescence / fizzes (1) negative electrode: turns pink (1) electrolyte: blue colour (of solution) fades (1)	**3**
3(c)(ii)	inert / (relatively) unreactive	**1**
3(d)	$Cu^{2+}(aq) + Mg(s) \rightarrow Cu(s) + Mg^{2+}(aq)$ correct formulae and balancing (1) correct state symbols – dependent on correct formulae (1)	**2**
3(e)	mass of copper = 2.00 g (1) mol copper = $\frac{2.00}{64}$ **and** mol oxygen = $\frac{0.25}{16}$ **OR** mol copper = 0.0313 **and** mol oxygen = 0.0156 (1) empirical formula is Cu_2O (1)	**3**

(b) Aqueous ammonia is added dropwise until in excess to a small volume of aqueous zinc chloride.

Describe the observations during this addition.

....

Question	Answer	Marks
8(f)(i)	bacterial oxidation / reaction with (acidified) potassium manganate(VII)	**1**
8(f)(ii)	sodium <u>ethanoate</u> and water	**1**

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>9(a)</td><td>(chemically) combined with water</td><td>1</td></tr>
<tr><td>9(b)</td><td>white precipitate (1)

soluble in excess resulting in colourless solution (1)</td><td>2</td></tr>
<tr><td>9(c)</td><td>white precipitate</td><td>1</td></tr>
<tr><td>9(d)</td><td><table>
<tr><td>concentration</td><td>dilute</td><td>concentrated</td></tr>
<tr><td>anode</td><td>oxygen (and water) / O_2 (and H_2O) (1)</td><td>chlorine / Cl_2 (1)</td></tr>
<tr><td>cathode</td><td>hydrogen / H_2 (1)</td><td>hydrogen / H_2 (1)</td></tr>
</table></td><td>4</td></tr>
<tr><td>9(e)</td><td>ions can move in aqueous / ions cannot move in solid</td><td>1</td></tr>
</table>

(b) Describe a chemical test for aluminium ions.

....

....

....

.... [2]

(c) The aqueous aluminium sulfate formed is crystallised to make hydrated aluminium sulfate,

Question	Answer	Marks
6(a)	$2Al(s) + 6H^+(aq) \rightarrow 2Al^{3+}(aq) + 3H_2(g)$ balancing (1) correct state symbol dependent on formulae (1)	2
6(b)	add (aqueous) sodium hydroxide (1) white ppt that redissolves / white ppt. soluble in excess (1) **OR** add (aqueous) ammonia (1) white precipitate (1)	2
6(c)	M_r for aluminium sulfate is 342 (1) x = 18 (1)	2
6(d)	**mixture** of a metal with another element	1

Question	Answer	Marks
7(a)	global warming / greenhouse effect / climate change	1
7(b)	acid rain / damage to buildings / killing living things	1
7(c)(i)	to increase the surface area (1) more exposed particles / more collisions per second (1)	2
7(c)(ii)	**particles** move slower / **particles** have less **kinetic** energy (1) fewer successful collisions / fewer collisions above that of activation energy (1)	2
7(d)(i)	(sulfur or sulfur dioxide) gains oxygen	1

(c) A sample of sodium chloride is tested using a flame test.

State the colour of the flame seen in this test.

.... [1]

Question	Answer	Marks
2(b)(iii)	**Any three from:** (potassium) floats on water / moves across the surface of water (1) (potassium) melts (1) bubbles / fizzes / effervesces (1) forms a colourless solution / piece gets smaller and smaller (1) lilac flame (1)	3
2(c)	yellow / orange (1)	1

Question	Answer	Marks
3(a)(i)	(moles of zinc oxide =) 3.50 / 81 **OR** 0.0432 (1) (moles of HCl =) 0.0600 (1) idea that need 0.0864 moles of HCl / only 0.0300 moles of ZnO react (1)	3
3(a)(ii)	so **all** the **acid** is used up (1)	1
3(a)(iii)	filtration (1)	1
3(b)	barium nitrate or barium chloride **AND** any soluble sulfate (1)	1
3(c)(i)	nitric acid **and** sodium hydroxide (1)	1
3(c)(ii)	titration (1)	1

(ii) an atom that forms an ion that gives a lilac colour in a flame test

.... [1]

(iii) an atom of a monatomic gas

Question	Answer	Marks
1(a)(i)	**G**	**1**
1(a)(ii)	**H**	**1**
1(a)(iii)	**F**	**1**
1(a)(iv)	**E**	**1**
1(a)(v)	**D**	**1**
1(b)	(number of protons) 24 (1) (number of neutrons) 29 (1)	**2**

Question	Answer	Marks
2(a)(i)	oxidation **and** reduction occur **simultaneously**	**1**
2(a)(ii)	CO removes oxygen **from** iron oxide / CO gains oxygen **from** the iron oxide	**1**

2 You are provided with solutions **R** and **S**.

(a) **(i)** Do the tests on **R** shown in the table.

Record your observations in the table.

test no.	test	observations
1	To 1 cm depth of **R** in a test-tube, add aqueous sodium hydroxide drop-by-drop until a change is seen. Add excess aqueous sodium hydroxide.	
2	To 1 cm depth of **R** in a test-tube, add aqueous ammonia drop-by-drop until a change is seen. Add excess aqueous ammonia.	
3	To 1 cm depth of **R** in a test-tube, add 3 drops of dilute nitric acid and then add 1 cm depth of aqueous silver nitrate.	

[6]

Conclusions

(ii) Name the cation in **R**.

cation .... [1]

(iii) Name the anion in **R**.

anion .... [1]

Question	Answer	Marks
	General points **R** is chromium(III) chloride **S** is iron(II) sulfate For gases: to gain credit for the name of the gas produced, the test must be at least partially correct. Solutions: colourless is not equivalent to clear and clear is not equivalent to colourless No credit is given for conclusions based upon incorrect observations.	
2(a)(i)	(test 1) (grey-)green ppt (1) dissolves (in excess) (1) green solution (1) (test 2) (grey-)green ppt (1) insoluble (in excess) (1) (test 3) white ppt (1)	**6**
2(a)(ii)	(cation is) chromium(III)	**1**
2(a)(iii)	(anion is) chloride	**1**

(a) Do a flame test on **one** sample of **A**. Describe the method you use.

method ....

....

....

observations ....

....

conclusion ....

....

[4]

(b) **(i)** Place the other sample of **A** in a test-tube and add 3 cm depth of dilute hydrochloric acid.

Question	Answer	Marks
2(a)	**M1** dip wire / splint / rod into solid **M2** put wire / splint into blue / roaring / airhole open / non-luminous / colourless flame **M3** blue – Green colour in flame. **M4** copper ions / Cu^{2+} present.	**4**
2(b)(i)	**M1** effervescence / fizzing / bubbling. **M2** (bas turns) limewater milky / cloudy. **M3** carbon dioxide / CO_2 formed. **M4** blue-green solution (formed). **M5** carbonate (ions) / CO_3^{2-} present	**5**
2(b)(ii)	no solid remains undissolved / effervescence stops	**1**
2(c)(i)	**M1** blue precipitate **M2** (forms) deep blue solution (in excess)	**2**
2(c)(ii)	blue precipitate (insoluble in excess)	**1**
2(d)	copper carbonate, $CuCO_3$	**1**
2(e)	**M1** chloride white precipitate **and** bromide cream / off white precipitate **and** iodide (pale) yellow precipitate. **M2** result for **B** off white / cream precipitate **M3** bromide / Br^- ions present	**3**